



Exp : 14 Write the Python Program to implement Alpha & Beta pruning algorithm for gaming

Aim:

To write a python program for vacuum cleaner problem.

Algorithm:

Step1: Start at the root node with initial values of Alpha as negative infinity and Beta as positive infinity.

Step2: If the current node is a leaf node or reaches the target depth, return its utility score.

Step3: If it is the Maximizer's turn, recursively evaluate child nodes to find the maximum score, updating Alpha with the highest value found so far.

Step4: If it is the Minimizer's turn, recursively evaluate child nodes to find the minimum score, updating Beta with the lowest value found so far.

Step5: At any node, if Beta becomes less than or equal to Alpha, prune the remaining unexamined child branches of that node and stop evaluating them.

Step6: Return the optimal score to the root node after evaluating only the unpruned branches.

Program:

```
import math

def alpha_beta_pruning(depth, node_index, is_max, values, alpha, beta, target_depth):
    if depth == target_depth:
        return values[node_index]

    if is_max:
        best = -math.inf
        for i in range(2):
            val = alpha_beta_pruning(depth + 1, node_index * 2 + i, False, values, alpha, beta, target_depth)
            best = max(best, val)
```

```

        alpha = max(alpha, best)

        if beta <= alpha:
            break

    return best

else:
    best = math.inf

    for i in range(2):
        val = alpha_beta_pruning(depth + 1, node_index * 2 + i, True, values, alpha, beta,
target_depth)

        best = min(best, val)

        beta = min(beta, best)

        if beta <= alpha:
            break

    return best

if __name__ == '__main__':
    leaf_scores = [3, 5, 6, 9, 1, 2, 0, -1]

    tree_depth = int(math.log2(len(leaf_scores)))

    optimal_value = alpha_beta_pruning(0, 0, True, leaf_scores, -math.inf, math.inf,
tree_depth)

    print("Leaf node scores:", leaf_scores)

    print("The optimal value for Maximizer with pruning is:", optimal_value)

```

OUTPUT:

```

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Leaf node scores: [3, 5, 6, 9, 1, 2, 0, -1]
The optimal value for Maximizer with pruning is: 5

```

Result: Thus, the program was executed and found that the output was found to be correct.